

Greywater, Reed Bed, Pond

Carpathian OER Commons — Utilities Module

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Maturity: L1 — designed and costed, untested. Every cost line is an estimate or a listed price, labelled as such in Section 6; several quantities are assumed until the walk and a percolation test replace them. Nothing has been built. Read every figure below at this level and no higher.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the sanitation working paper, EN v0.2, Sections 6, 7, 8.5, and 8.6. Three things change from the paper, decided on 2026-09-29: the reed bed is sized at 5 m² a person, not 4; every used water settles in the septic tank before the bed, not only the toilets'; and the bed drains to a soakaway, with the paper's pond kept as the option where the authorities allow it. The paper is sized at 25 people; every quantity here is recalculated at 34.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Every used water on the site — showers, basins, the kitchen, the two existing flush toilets, and the rinse of the toilet buckets — made clean enough to return to the ground: a grease trap, a septic tank, a planted gravel bed, and a soakaway, all by gravity, working under snow.

Five rules. Nothing raw reaches the bed: the kitchen passes a grease trap, and all used water settles in the septic tank first. Below the surface: the water in the bed stays 5 to 15 cm under the gravel, so there is no standing water, little smell, and nothing to reach. Gravity before pumps: the chain runs downhill, from building to tank to bed to soakaway. Leave the reeds standing: they are cut in spring, not autumn, because their litter keeps the bed working in winter. The ground decides the outflow: the soakaway is sized by a percolation test, and the pond is built only where the authorities allow it and its distances are met.

Connections. Depends on: the water module, whose used water this is; the toilets module, whose two flush toilets carry about a third of uses; the terra preta module, whose bucket rinse goes to the septic tank; the washroom block and the kitchen and dining modules, for the showers, basins, and kitchen; the Phase 1 walk, with the fall of the ground, the depth to groundwater, a percolation test, and every distance in Section 4 measured; the written enquiries in Section 10. Feeds: the ground, through the soakaway; or, where permitted, the pond.

Patterns. Grows from: not yet named. Where the valley let its used water go, and whether any wet ground was kept for it, are questions for the elders. Feeds: not yet named. Resonates with: 104 Site Repair — tank, bed, and soakaway on the poorer ground, below the buildings; 172 Garden Growing Wild — the reed bed as a wild edge, left to itself but for its spring cut; 25 Access to Water — every part of it outside the belt along the stream and the pond. In tension with: 64 Pools and Streams — the soakaway returns the water out of sight, where the paper's pond would have left a visible end; 71 Still Water — the pond, if it is built, is a treatment pond, fenced and shallow at its edges, never a place to swim.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; the bed works under snow
Altitude	[set]
Group size	capacity: 34 people, the example below
Flow	about 2.2 m ³ a day at capacity
Fall	building above tank above bed above soakaway
Soil	permeable, 0.1 m a day or more, for the soakaway
Groundwater	at least 1 m below the soakaway's base, [set]
Sanitation	dry toilets for two-thirds of uses

Where the soil does not take water — clay, rock near the surface, or high groundwater — the soakaway does not work, and the outflow is the pond with its permit, or a sealed alternative the authorities name.

3. Sizing method

What you must measure yourself before using this module:

- **Your flow** — the water used on a full day at capacity, from the water meter or the cistern's fall, and the kitchen's share of it.
- **Your fall** — the levels of building, tank, bed, and soakaway, from the walk.
- **Your soil** — a percolation test where the soakaway would go, and the depth to groundwater in the wettest month.

- **Your distances** — as in Section 4, measured with a tape.

Our example: capacity, 34 people; greywater_per_person, 55 L a day, [assumed]; toilet_uses, 5 a day; bucket_share, two-thirds; flush_volume, 5 L, [assumed]; meals_per_day, 3, [assumed]; reed_bed_area, 5 m² a person; soil_loading, 16 to 24 L a m² a day. Values from the site parameters file, held in the workshop.

3.1 The flow

Stream	Calculation	Our example
Showers, basins, kitchen	$34 \times 55 \text{ L}$	about 1,870 L
Flush toilets, uses	$34 \times 5 \times \text{one-third}$	about 57
Flush toilets, water	$57 \times 5 \text{ L}$	about 283 L
Bucket rinse	the terra preta module	[set]
A day at capacity	sum	about 2.2 m³

The water module sizes the spring for 75 L a person a day, 2.55 m³ at capacity; about 2.2 m³ of it returns as used water. Greywater from basic, piped settings runs from 20 to 30 L a person a day where water is scarce to 90 to 120 where it is not (Morel and Diener, Greywater Management in Low and Middle-Income Countries, Eawag, 2006); the paper's 55 L is kept, [assumed], until the meter replaces it.

3.2 The parts

Part	Calculation	Our example
Grease trap	EN 1825-2, meal method	nominal size about 0.5
Septic tank	$3 \text{ days} \times 2.2 \text{ m}^3$	about 6.5 m ³
First chamber	0.75 of the volume	about 4.9 m ³
Second chamber	0.25 of the volume	about 1.6 m ³
Reed bed	$34 \times 5 \text{ m}^2$	170 m ²
Gravel in the bed	$170 \text{ m}^2 \times 0.7 \text{ m}$	about 119 m ³
Soakaway base	$2.2 \text{ m}^3 \div 16\text{--}24 \text{ L a m}^2$	about 88–132 m ²
Soakaway trench, 0.6 m wide	$\text{base} \div 0.6 \text{ m}$	about 147–220 m

The grease trap. EN 1825-2 sizes a grease separator from meals, a volume a meal, and a peak factor; with the works-canteen values of 5 L a meal and a factor of 20, as given in a manufacturer's sizing sheet (Zehnder), 102 meals a day over 8 hours give a flow of about 0.35 L a second, and a nominal size of about 0.5 where the water is under 60 °C. The kitchen's own count sets it, [set].

The septic tank. ДБН B.2.5-75:2013, clause 10.8.3: septic tanks for flows up to 20 m³ a day; a volume of at least three days' flow up to 5 m³ a day; two chambers up to 10 m³ a day, divided 0.75 and 0.25. The paper's tank took the flush toilets only; here all used water settles before the bed, because primary treatment is essential to a subsurface bed (Tilley and colleagues, Compendium of Sanitation Systems and Technologies, 2014), and greywater carries hair, lint, and fat that would clog its inlet.

The reed bed. Horizontal subsurface flow, lined. DWA-A 262 (2017) gives 5 m² a person for horizontal-flow beds, at least 20 m², and a filter layer of at least 50 cm, as reported by German planning sources; Tilley and colleagues give 5 to 10 m². The paper's 4 m² a person is below both; 5 m² is used. Saturated depth about 0.6 m, with 0.1 m of dry gravel above the water; gravel of 5 to 10 mm in the bed and 50 to 100 mm at the inlet and outlet zones (Wafler, Small-Scale Constructed Wetlands, seecon, 2008); the water level held 5 to 15 cm below the surface by an adjustable outlet; a liner of at least 1 mm (DWA-A 262); reeds planted at about four a m² in early spring (UN-Habitat, Constructed Wetlands Manual, 2008).

The soakaway. Filtration trenches take up to 15 m³ a day, in sandy, sandy-loam, or loamy soil taking at least 0.1 m a day (ДБН B.2.5-75:2013, clause 10.8.2). The loading is taken from a soils table for septic-tank effluent: 0.4 to 0.6 US gallons a square foot a day for loam, sandy loam, and silt loam, by structure (Tyler, Hydraulic Wastewater Loading Rates to Soil, University of Wisconsin), about 16 to 24 L a m² a day. The bed's outflow is cleaner than septic effluent and the same table allows it more; that allowance is not used here. The percolation test replaces the range; the upper length is costed.

Winter. A subsurface bed keeps working under snow; beds without snow cover have frozen and failed in a severe winter, and 15 cm of mulch over a 5 cm air gap kept one working at -28 °C (Wallace, Parkin, and Cross, Water Science and Technology, 2001). The reed litter is left standing as the paper holds; a mulch is added if the first winter shows the bed freezing, [set].

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Grease trap, about 0.5 L a second	1	a larger trap
Septic tank, two chambers	about 6.5 m ³	concrete rings; plastic

Item	Quantity	Alternatives in the design
Distribution chamber, pipework	1 set	none named
Liner, 1 mm or more	about 204 m ²	HDPE; EPDM
Geotextile, under and over the liner	about 408 m ²	none named
Gravel, 5–10 mm, coarse at the ends	about 119 m ³	none named
Inlet and outlet manifolds	1 set	none named
Adjustable outlet, inspection points	1 set	a turned-down elbow
Common reed	about 680 plants	from the valley
Soakaway gravel	about 66 m ³	none named
Soakaway pipe, 110 mm, drilled	about 220 m	drainage pipe
Geotextile over the soakaway	about 220 m ²	none named

The siting, as designed. Every distance below is the larger of what the instruments in Section 10 ask, and every one is measured on the ground before anything is dug; none has been measured yet.

From	To	At least
The septic tank	any building	5 m
The soakaway	any dwelling	25 m
The reed bed	any dwelling	25 m, [set]
Every part	the spring, never upslope	50 m
Every part	the stream and a pond, slope	50 m
Every part	places where children are	20 m
The pond, if built	any dwelling	up to 200 m

No distance is set for a reed bed in the code found; the soakaway's 25 m is used until the enquiry answers. The soakaway also stays outside the stream's water protection zone, which is wider than the strip, and in which filtration fields are not allowed (Section 10). If the pond counts as a biological pond under the code, its distance from housing is 200 m.

The layout, as designed. Building, grease trap, septic tank, distribution chamber, reed bed, and soakaway in a line down the fall, each below the last; the reed bed on poorer ground, lined, level across, fed across its full width; the soakaway downslope of everything and upslope of nothing that is drunk. Drawn as relationships only; the walk sets the rest.

The pond, as the option. The paper's pond: about 25 m², shallow, lined, planted, fenced with a gate, shallow at its edges, taking only the bed's outflow. A pond of this kind gives little treatment in the cold months and needs mosquito control (SSWM, Waste Stabilisation Ponds); it is a habitat and a visible end, not a treatment step, and is built only where the distance and the permit in Section 10 are met. The paper costed it at 37,200 UAH at 25 people — liner, excavation, fence and gate — outside the total below.

5. Build

Step	Trade	Time	If done out of order
Enquiries answered	household	[from L2]	permit refused
Walk, levels, percolation	household	[from L2]	soakaway fails
Excavation, tank, bed, trench	excavator	[from L2]	[from L2]
Septic tank and grease trap	builder	[from L2]	[from L2]
Bed lined, gravel laid	builder	[from L2]	liner torn
Manifolds, outlet, pipework	plumber	[from L2]	short-circuit flow
Soakaway laid, covered	builder	[from L2]	[from L2]
Reeds planted, early spring	household	[from L2]	poor rooting
Filled, flow checked	plumber	[from L2]	ponding

6. Cost

Estimates, not quotations: each line is either a price listed by a Ukrainian retailer or service, seen on 2026-09-29, or the sanitation paper's own estimate for autumn 2026, marked "paper". Place: not a quotation; western Ukrainian listings where found, national otherwise. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Pre-treatment

Item	UAH	EUR	Basis
Grease trap, 0.5 L a second	4,876	94	estimated, listing
Septic tank, about 6.5 m ³	97,500	1,875	paper, scaled
Distribution chamber, pipework	10,000	192	paper
Subtotal	112,376	2,161	

The septic tank is the paper's 45,000 UAH for 3 m³, scaled to 6.5 m³, [assumed]; a 3,000 L plastic tank is listed at 30,330 UAH, and a concrete ring of 1.5 m at 4,075 UAH, for a quotation to replace it.

6.2 Reed bed, 170 m²

Item	UAH	EUR	Basis
Liner, 204 m ² , paper's rate	102,000	1,962	paper, scaled
Geotextile, 408 m ² at 46	18,768	361	estimated, listing
Gravel, 119 m ³ delivered, 1,100	130,900	2,517	estimated, listing
Manifolds, pipes, inspection points	25,500	490	paper, scaled
Subtotal	277,168	5,330	

The liner is the paper's 60,000 UAH for 120 m², the same 500 UAH a m² at 204 m², the paper's ratio of liner to bed, [assumed]; a 1 mm HDPE geomembrane is listed at 134.62 UAH a m² and a 0.8 mm EPDM at 492, with the fitting not included. The manifolds are the paper's 15,000 UAH scaled to the bed's area, [assumed]. Reeds from the valley are counted as contribution; bought, 680 plants at a listed 90 UAH would be 61,200 UAH.

6.3 Soakaway

Item	UAH	EUR	Basis
Gravel, 66 m ³ delivered at 1,100	72,600	1,396	estimated, listing
Pipe, 74 × 3 m of 110 mm at 573	42,402	815	estimated, listing
Geotextile, 220 m ² at 46	10,120	195	estimated, listing

Item	UAH	EUR	Basis
Subtotal	125,122	2,406	

6.4 Earthworks

Item	UAH	EUR	Basis
Excavator, 30 hours at 1,187	35,610	685	listing; hours assumed
Subtotal	35,610	685	

6.5 Summary

Section	UAH	EUR
Pre-treatment	112,376	2,161
Reed bed	277,168	5,330
Soakaway	125,122	2,406
Earthworks	35,610	685
Construction subtotal	550,276	10,582
Design and sanitary review, 8 percent	44,000	846
Contingency, 20 percent	110,100	2,117
Total external cost	704,376	13,546

Against the paper's 281,800 UAH for pre-treatment, reed bed, and pond at 25 people, the difference is the larger bed, the septic tank for all used water, the soakaway, and gravel priced delivered. The soakaway is costed at its upper length, 220 m; the percolation test may halve it. The percentages are the paper's own.

Contribution, not in the total. Common reed from the valley, with the landowner's leave, planted by paid hands.

Running cost. The septic tank desludged every two to five years (Tilley and colleagues, 2014), at a listed 4,600 UAH a trip for up to 4 m³ of sludge. The grease trap cleaned weekly. The reeds cut in spring. The work is paid, [set] hours.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2].

- **Weekly:** the grease trap cleaned, its fat to the terra preta vessels in small measure.
- **Monthly:** the inspection points checked for water at the surface; the outlet level set.
- **Quarterly:** the septic tank inspected; its scum and sludge measured.
- **Spring:** the reeds cut, never in autumn; the soakaway's ground walked for wet patches.
- **Every two to five years:** the septic tank desludged.

How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Bed inlet clogs	water at the surface	clean the inlet zone
Fat in the bed	grey film at the inlet	clean the trap weekly
Bed freezes	no outflow in frost	mulch, 15 cm over air
Septic tank full	scum at the outlet	desludge, 4,600 UAH
Soakaway saturates	wet ground, smell	longer trench; rest it
Liner punctured	bed runs dry	[from L2]
Outflow refused	written answer	the pond with a permit

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-29, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-29 and read on secondary copies and official summaries, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **ДБН В.2.5-75:2013**, sewerage, external networks and structures, with Amendment 1: clause 10.8.3, septic tanks as in Section 3; clause 10.8.2, soil treatment only in permeable soil and up to 15 m³ a day; clause 3.6, bioengineering structures using higher aquatic plants as a biofilter are named; Table 30 with its notes: septic tanks 5 m and filtration trenches 25 m from housing, biological ponds 200 m for the smallest plants. No distance is given for a reed bed.
- **Water Code, No. 213/95-BP**, Arts. 48 and 49: discharge of pollutants to a water body is special water use, needing a permit from the state water agency's regional office, with no exemption by volume. Art. 70: discharge only within set limits, and never by way of ravines or low ground. Art. 87: no filtration fields in a water protection zone. Arts. 88–89: a protective strip of 25 m, 50 m on a slope, along small streams and ponds. Whether infiltration to ground counts as discharge to a water body is not verified.
- **Cabinet Resolution No. 321 of 13.03.2002**, the permit procedure for special water use: a first permit for discharge is short-term, three years, with limits calculated for each outlet.
- **Cabinet Resolution No. 1100 of 11.09.1996**, as amended 10.12.2025: limits on discharge set for each water user, with no threshold for small dischargers.
- **ДСанПіН 2.2.4-171-10**, point 3.32: a spring capture at least 50 m from sewerage structures.
- **МОХ Order No. 145 of 17.03.2011**, point 2.22: cesspits at least 20 m from dwellings and places where children play; taken here for the septic tank and soakaway as the larger of the two figures.

The written enquiries: to the regional office of the state water agency, whether the soakaway or the pond needs a special water-use permit and limits; to the district sanitary service and the planning authority, the distance for a reed bed and whether the pond counts as a biological pond; to the council, the stream's protective strip and water protection zone at the site.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley:** less greywater; the bed and soakaway shrink with the flow; keep the reeds watered in drought.
 - **A colder valley:** mulch the bed from the first winter; lay the tank and pipes below the frost line.
 - **A smaller group:** the bed at 5 m² a person, never below 20 m²; the septic tank at three days' flow.
 - **A larger group:** above 10 m³ a day, three septic chambers; above 15 m³, no soakaway under the code found here.
 - **Flush toilets throughout:** the flow rises; recalculate the tank, the bed, and the soakaway.
 - **Impermeable soil:** no soakaway; the pond with its permit, or what the authorities name.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** the sanitation working paper, EN v0.2; DWA-A 262, 2017, as reported by German planning sources; Tilley and colleagues, Compendium of Sanitation Systems and Technologies, Eawag, 2014; UN-Habitat, Constructed Wetlands Manual, 2008; Wafler, Small-Scale Constructed Wetlands, seecon, 2008; Wallace, Parkin, and Cross, 2001, on cold-climate wetlands; Morel and Diener, Eawag, 2006, on greywater; EN 1825-2 as applied in a manufacturer's sizing sheet; Tyler, University of Wisconsin, on soil loading; SSWM on ponds; ДБН Б.2.5-75:2013; Ukrainian retail and service listings of 2026-09-29 for prices.
- **Traditional knowledge:** none. Where the valley let its used water go is for the elders.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** [hardware licence — open decision]
- **Measurement data:** none yet
- **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.1 may be published as it stands, at L1, in English, Ukrainian, and German
- **Wartime review:** 2026-09-29, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; the distances are the instruments' minimums, not measurements of the site; accepted

13. What to check

- **The enquiries** — the water-use permit, the reed bed's distance, the pond's status, the strip and the water protection zone.
 - **The flow** — greywater_per_person and flush_volume from the meter.
 - **The fall and the groundwater** — from the walk, in the wettest month.
 - **The percolation test** — replacing soil_loading and the soakaway's length.
 - **The distances** — every row of the siting table, measured with a tape.
 - **The grease trap** — sized from the kitchen's meals.
 - **The septic tank** — a quotation replacing the scaled estimate.
 - **The liner** — material chosen, fitting quoted; its area from the layout.
 - **The excavator hours** — from a quotation.
 - **The DWA-A 262 values** — read in the standard itself.
 - **The Ukrainian row** — every instrument read and signed.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.
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